

Assignment 3

00103335: Deep Learning and Reinforcement Learning

Fall 2026

Problem 1

Write $v_t = \mathbb{E}|\delta_t|^4$ and $q_t = 1 - \lambda\eta_t \in [0, 1]$. Conditional Hölder gives, for $r = 2, 3$,

$$\mathbb{E}[|\xi_t|^r \mid \mathcal{F}_t] \leq (\mathbb{E}[|\xi_t|^4 \mid \mathcal{F}_t])^{r/4} \leq \sigma^r.$$

Moreover, $|u + v|^4 \leq 8(|u|^4 + |v|^4)$ yields $v_{t+1} \leq 8v_t + 8\eta_t^4\sigma^4$, so every v_t is finite by induction.

(a) Expanding conditionally and using $\mathbb{E}[\xi_t \mid \mathcal{F}_t] = 0$,

$$\begin{aligned} \mathbb{E}[|\delta_{t+1}|^4 \mid \mathcal{F}_t] &= q_t^4\delta_t^4 + 6q_t^2\eta_t^2\delta_t^2\mathbb{E}[\xi_t^2 \mid \mathcal{F}_t] \\ &\quad + 4q_t\eta_t^3\delta_t\mathbb{E}[\xi_t^3 \mid \mathcal{F}_t] + \eta_t^4\mathbb{E}[\xi_t^4 \mid \mathcal{F}_t] \\ &\leq q_t^4\delta_t^4 + 6\sigma^2\eta_t^2\delta_t^2 + 4\sigma^3\eta_t^3|\delta_t| + \sigma^4\eta_t^4. \end{aligned}$$

The inequality $2uv \leq u^2 + v^2$ gives

$$4\sigma^3\eta_t^3|\delta_t| \leq 2\sigma^2\eta_t^2\delta_t^2 + 2\sigma^4\eta_t^4.$$

For any $\varepsilon > 0$, completing the square gives

$$8\sigma^2\eta_t^2\delta_t^2 \leq \varepsilon\eta_t\delta_t^4 + \frac{16\sigma^4}{\varepsilon}\eta_t^3.$$

Fix $c \in (0, 4\lambda)$ and set $\varepsilon = (4\lambda - c)/2$. For $0 \leq x \leq 1$, $(1 - x)^4 \leq 1 - 4x + 6x^2$. Since $\eta_t \rightarrow 0$, eventually $\eta_t \leq 1$ and $6\lambda^2\eta_t \leq \varepsilon$. Hence

$$q_t^4 + \varepsilon\eta_t \leq 1 - (4\lambda - \varepsilon)\eta_t + 6\lambda^2\eta_t^2 \leq 1 - c\eta_t.$$

Taking expectations and using $\eta_t^4 \leq \eta_t^3$ proves

$$\boxed{v_{t+1} \leq (1 - c\eta_t)v_t + C_0\eta_t^3}, \quad C_0 = \left(\frac{16}{\varepsilon} + 3\right)\sigma^4.$$

(b) Define the relative decrease on the scale η_t by

$$D_t = \frac{\eta_t^2 - \eta_{t+1}^2}{\eta_t^3} = \frac{(t + t_0)^\alpha}{a} \left[1 - \left(1 + \frac{1}{t + t_0} \right)^{-2\alpha} \right].$$

With $s = t + t_0$, the mean value theorem gives $1 - (1 + 1/s)^{-2\alpha} = (2\alpha/s)(1 + \zeta_s)^{-2\alpha-1}$ for some $0 < \zeta_s < 1/s$. Thus

$$D_t \longrightarrow d_* := \begin{cases} 0, & \alpha < 1, \\ 2/a, & \alpha = 1. \end{cases}$$

In both cases $d_* < 4\lambda$: for $\alpha = 1$ this is exactly $2\lambda a > 1$. Choose $d_* < d < c < 4\lambda$ and apply (a) with this c . For some N , all $t \geq N$ satisfy its recursion, $D_t \leq d$, and $c\eta_t \leq 1$. Choose a finite constant

$$C \geq \max \left\{ \frac{C_0}{c-d}, \max_{1 \leq t \leq N} \frac{v_t}{\eta_t^2} \right\}.$$

If $v_t \leq C\eta_t^2$ for $t \geq N$, then

$$\begin{aligned} v_{t+1} &\leq (1 - c\eta_t)C\eta_t^2 + C_0\eta_t^3 \\ &\leq C\eta_t^2 - Cd\eta_t^3 \leq C(\eta_t^2 - D_t\eta_t^3) = C\eta_{t+1}^2. \end{aligned}$$

Induction, together with the choice of C for $t \leq N$, proves $\boxed{\mathbb{E}|\delta_t|^4 \leq C\eta_t^2}$ for every t .

(c) For $M > 0$, Markov's inequality and (b) give

$$\mathbb{P}(|\delta_t| > M\sqrt{\eta_t}) \leq \frac{\mathbb{E}|\delta_t|^4}{M^4\eta_t^2} \leq \frac{C}{M^4}.$$

For every $\epsilon > 0$, choose M such that $C/M^4 < \epsilon$. This proves $\delta_t = O_P(\sqrt{\eta_t})$.

(d) Since $2\alpha > 1$, $\sum_{t \geq 1} \eta_t^2 < \infty$. For any integer $k \geq 1$,

$$\sum_{t=1}^{\infty} \mathbb{P}(|\delta_t| > 1/k) \leq k^4 \sum_{t=1}^{\infty} v_t \leq Ck^4 \sum_{t=1}^{\infty} \eta_t^2 < \infty.$$

The first Borel–Cantelli lemma implies $|\delta_t| \leq 1/k$ eventually almost surely. Intersecting these probability-one events over k proves $\boxed{\delta_t \rightarrow 0 \text{ almost surely}}$. No independence is needed.

Problem 2

Let $e_t = \theta_t - \theta^*$ and $\Delta_t = \Phi(\theta_t) - \Phi(\theta^*) \geq 0$. All quantities below are integrable: the recursion and $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$ give finite $\mathbb{E}\|e_t\|^2$ by induction. Conditional Jensen gives $\|\nabla\Phi(\theta_t)\|^2 \leq \mathbb{E}[\|g_t\|^2 | \mathcal{F}_t] \leq G^2$. Convexity then implies $0 \leq \Delta_t \leq G\|e_t\|$.

(a) Expand the squared distance and condition on $\mathcal{F}_t$:

$$\begin{aligned} \mathbb{E}[\|e_{t+1}\|^2 | \mathcal{F}_t] &= \|e_t\|^2 - 2\eta_t e_t^\top \mathbb{E}[g_t | \mathcal{F}_t] + \eta_t^2 \mathbb{E}[\|g_t\|^2 | \mathcal{F}_t] \\ &\leq \|e_t\|^2 - 2\eta_t e_t^\top \nabla\Phi(\theta_t) + \eta_t^2 G^2. \end{aligned}$$

Convexity at θ_t gives

$$\Phi(\theta^*) \geq \Phi(\theta_t) + \nabla\Phi(\theta_t)^\top (\theta^* - \theta_t), \quad e_t^\top \nabla\Phi(\theta_t) \geq \Delta_t.$$

Since $\eta_t > 0$, substitution gives

$$\boxed{\mathbb{E}[\|e_{t+1}\|^2 | \mathcal{F}_t] \leq \|e_t\|^2 - 2\eta_t \Delta_t + \eta_t^2 G^2.}$$

(b) Take expectations in (a), rearrange, and sum:

$$\begin{aligned} 2 \sum_{t=1}^T \eta_t \mathbb{E} \Delta_t &\leq \sum_{t=1}^T \left(\mathbb{E} \|e_t\|^2 - \mathbb{E} \|e_{t+1}\|^2 \right) + G^2 Q_T \\ &= \mathbb{E} \|e_1\|^2 - \mathbb{E} \|e_{T+1}\|^2 + G^2 Q_T \leq R^2 + G^2 Q_T. \end{aligned}$$

Since $\eta_t/S_T > 0$ and their sum is one, convexity gives, pathwise,

$$\Phi(\bar{\theta}_T^{(\eta)}) - \Phi(\theta^*) \leq \frac{1}{S_T} \sum_{t=1}^T \eta_t \Delta_t.$$

Taking expectations proves

$$\boxed{\mathbb{E}[\Phi(\bar{\theta}_T^{(\eta)})] - \Phi(\theta^*) \leq \frac{R^2 + G^2 Q_T}{2S_T}.}$$

(c) For a fixed horizon T , set $\eta_t = c/\sqrt{T}$. Then $S_T = c\sqrt{T}$, $Q_T = c^2$, and $\bar{\theta}_T^{(\eta)} = T^{-1} \sum_{t=1}^T \theta_t = \bar{\theta}_T$. The bound in (b) becomes

$$\mathbb{E}[\Phi(\bar{\theta}_T)] - \Phi(\theta^*) \leq \frac{1}{2\sqrt{T}} \left(\frac{R^2}{c} + G^2 c \right).$$

For $c > 0$,

$$\frac{R^2}{c} + G^2 c - 2RG = \left(\frac{R}{\sqrt{c}} - G\sqrt{c} \right)^2 \geq 0,$$

with equality exactly when $c = R/G$ (positive since $R, G > 0$). This choice gives

$$\boxed{\mathbb{E}[\Phi(\bar{\theta}_T)] - \Phi(\theta^*) \leq \frac{RG}{\sqrt{T}}.}$$